Design and Implementation of a Raspberry Pi-Based Mini Security Operations Center

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Introduction

Today, the exponential development of information technologies positively affects both industrial systems and our daily lives. However, it also creates many security vulnerabilities. Although corporations, government institutions and public organizations are able to protect their information systems more effectively due to having relatively better resources, the same situation does not apply to home networks. Especially with the development of IoT, methods such as reconnaissance and lateral movement can now be executed within home networks just as in corporate networks. Almost no users monitor the DNS traffic within their own network or the traffic coming to and going from their ports. Therefore, anomalies in home networks cannot be detected when there is no observation.

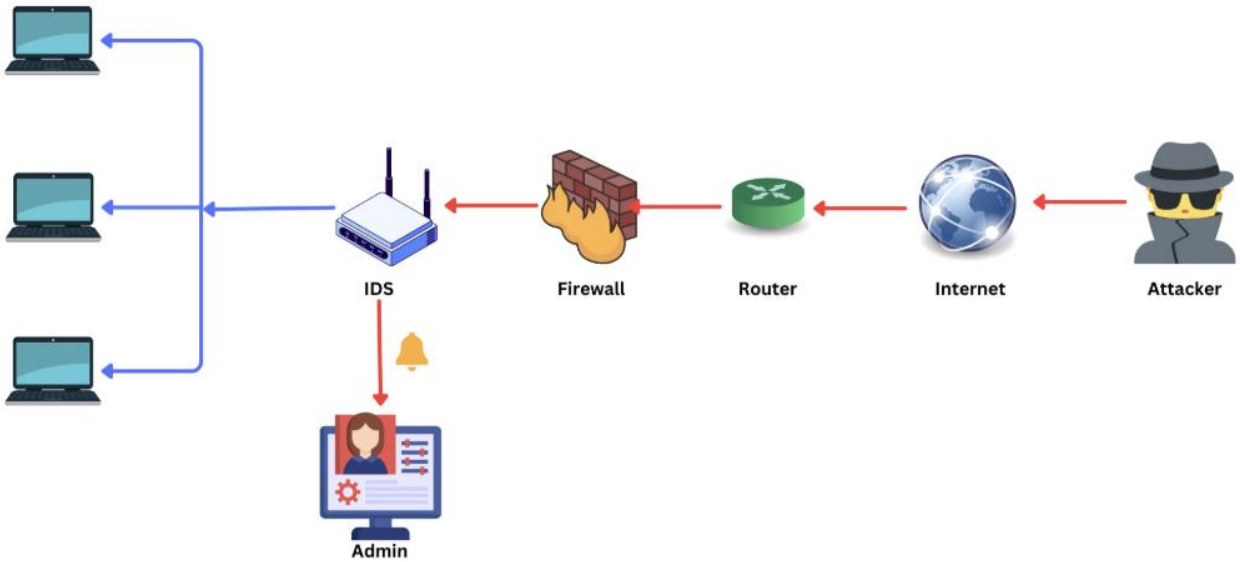
Corporations, with the resources they possess either establish their own network monitoring systems or purchase this service from companies that provide it. These companies carry out such activities through units called "Security Operations Center (SOC)". Corporate networks are continuously monitored, logged and classified 24/7 by real analysts. When observation is performed, anomalies can be detected and these anomalies can be prioritized according to their level of severity.

However, obtaining SOC services for a home network is quite costly and burdensome. However, this does not mean that the data of individuals connected to the home network are worthless or immune to attacks. For this reason, within the scope of this project a mini-SOC has been built that is both low-cost and capable of monitoring the entire home network. This project has been established as a Raspberry Pi 5-based system. This system performs port mirroring to detect DNS traffic on the home network and whether port scanning is being conducted. At the same time, it categorizes detected events according to their risk levels, assigns priorities and correlates multiple suspicious behaviors to increase the severity level of the alert. In order to automate these behaviors they have been converted into service-level processes meaning that as long as the device remains powered on, activities are regularly executed by the operating system. Additionally, all activities can be monitored through a dashboard installed on the terminal. Even if the user is not at the terminal the user can receive instant alerts via Gmail in the event of a possible alarm situation.

The fundamental question in this project is whether it is possible to establish a low-cost monitoring system for a home network without having the resources possessed by corporate companies. Within the scope of this project, the question of where privacy begins and where it ends has also been addressed. Corporations and government institutions can monitor the traffic coming to and going from the devices of their employees within their own networks, as this constitutes a professional activity rather than a private one. However, the situation is completely different for a home network. Since the connected devices are most likely personal devices and at this point personal privacy begins. Therefore, during the planning, construction, testing, and deployment phases of this project, all individuals connected to the home network were informed about the scope, purpose, and content of the monitored traffic, and their consent was obtained accordingly.

Background

The fundamental idea behind a network monitoring system is to observe network traffic, including incoming and outgoing connections, and detect potential anomalies. Rather than focusing on individual objects, such systems monitor activities and determine whether any behavior deviates from the normal flow of the network. One of the most important components of such systems is the Intrusion Detection System (IDS). The working principle of an IDS is relatively



Figur 1: IDS-based network monitoring architecture

straightforward. If an anomaly is detected within the normal traffic flow, the system generates an alert. Additionally, the IDS may compare current traffic patterns with previously identified attack signatures and trigger an alert if a match is found. Port scanning is generally associated with the reconnaissance phase of an attack. In this process, an attacker attempts to find a vulnerability to get inside of the system. Open ports represent potential attack surfaces just like doors, windows in a building. In corporate environments, port scanning attempts are typically detected by centralized security systems or SOCs whereas in home networks such behavior often goes unnoticed. Therefore, the detection of port scanning was covered by this project.

DNS (Domain Name System) traffic also plays an important role in analyzing network behavior. DNS queries indicate which domain names users or systems are attempting to access. Under normal conditions, DNS traffic follows certain patterns, such as “Apple.com” or “AWS”. However, sending an excessive number of queries within a short period, attempting to access unusual domain names or observing high DNS activity from a specific source may be considered anomalous behavior. In particular, DNS traffic can be abused in scenarios such as data exfiltration or command-and-control (C2) communication. For this reason, monitoring DNS traffic is an important data source for anomaly detection mechanisms. Therefore, DNS trafficking is also covered in this project.

However, there is a basic problem. Single events do not always produce reliable alerts. For example, a short-term traffic spike or a single connection attempt may not necessarily indicate a threat on its own. For this reason, event correlation plays an important role in Security Operations Center (SOC) environments. Correlation refers to the evaluation of multiple indicators from different sources together. If both abnormal DNS activity and port scanning behavior are observed from the same source, the confidence level of the event increases. This approach helps reduce false positives and generate more meaningful alerts. Therefore, combining anomaly detection with event correlation creates a more reliable monitoring mechanism. Just like in a real world, someone can knock our door just because they miss read the door number but if that person is looking after names this leads us to think this is not an accident this is by purpose.

System and Design Architecture

The system was implemented using the following hardware and software components:

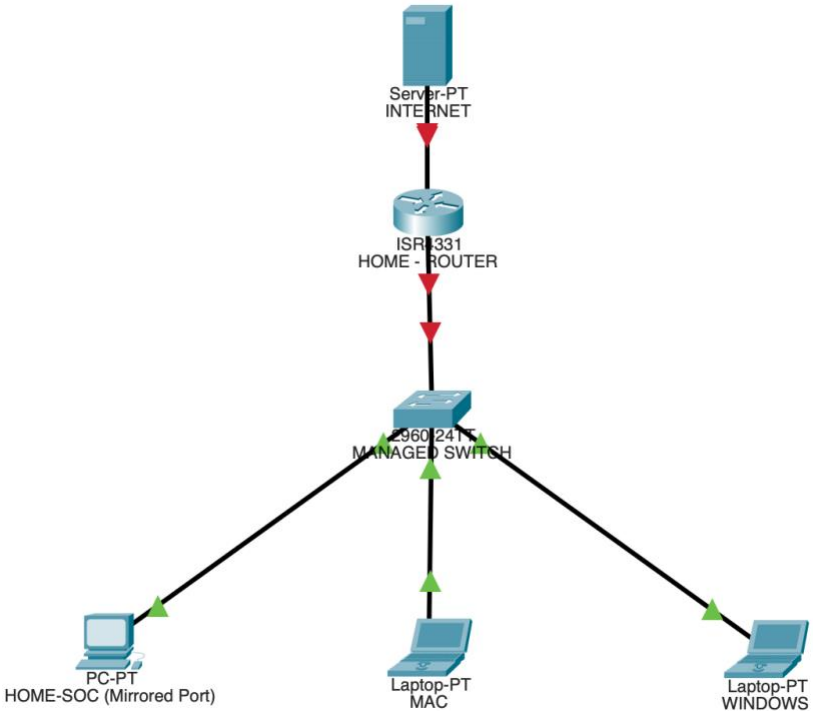
Hardware

- Raspberry Pi 5 (16GB RAM, 128GB disk)
- TP-Link TL-SG105E Managed Switch (Port Mirroring Enabled)
- Home Router (ISP-provided)
- Client Devices (macOS laptop, Windows laptop)

Software

- Ubuntu Desktop 24.04 LTS
- Python 3
- tcpdump
- systemd services
- Custom DNS detection script
- Custom port scan detection script
- Custom correlation engine
- Terminal-based dashboard (curses)
- Gmail SMTP alert integration

The system was designed as a lightweight and monitoring platform that could operate continuously within a home network environment. A Raspberry Pi 5 device running Ubuntu Desktop was selected as the core hardware component due to its low cost, low power consumption and sufficient processing capability for traffic monitoring tasks and lastly user-friendly UI. The device was connected to a managed switch supporting port mirroring, allowing passive observation of network traffic without actively interfering with communication.



Figur 2: Topology

Port mirroring was configured so that traffic from selected network ports was duplicated and forwarded to the Raspberry Pi interface. This configuration enabled the system to analyze DNS queries and connection attempts generated by devices within the home network. The monitoring architecture was intentionally kept passive to avoid modifying packets or introducing latency into the network.

The detection layer of the system consists of two independent anomaly detection modules: DNS anomaly detection and port scan detection. The DNS detection mechanism observes query volume within a defined time window and identifies bursts of activity that exceed predefined thresholds. These thresholds were selected to represent unusual behavior rather than normal browsing patterns. Similarly, the port scan detection module monitors rapid sequential connection attempts targeting multiple ports from the same source IP address. Such behavior is commonly associated with reconnaissance activity.

Each detected event is categorized according to a severity model consisting of four levels: INFO, MEDIUM, HIGH, and CRITICAL. Lower severity levels indicate minor anomalies, while higher levels represent behavior that may suggest coordinated or malicious activity. This classification structure was implemented to simulate common SOC prioritization practices and to make alert interpretation more structured.

To increase detection reliability, an event correlation mechanism was implemented. Instead of treating anomalies independently, the system evaluates whether multiple suspicious behaviors originate from the same source within a specified time window. For example, if abnormal DNS activity and port scanning behavior are detected from the same IP address, the system elevates the event to CRITICAL severity. This approach increases confidence in alerts and reduces the likelihood of reacting to isolated, non-malicious events.

All detection modules were implemented as system-level services using systemd. This design ensures that monitoring processes automatically start on boot and continue running without manual intervention by the user. Structured logging was implemented to record detection results in JSON format, enabling consistent event tracking and analysis. Log rotation and pruning mechanisms were also configured to prevent excessive disk usage.

In addition to backend detection mechanisms, a terminal-based dashboard was developed to provide real-time visibility into system activity. The dashboard displays categorized alerts and highlights critical events. Furthermore, an email notification mechanism was integrated to inform the user when high-severity alerts are generated, even if the dashboard is not actively monitored. This combination of passive monitoring, anomaly detection, correlation, and automated alerting forms the core architecture of the implemented mini-SOC system.

Practical Implementation and Results

To test whether the implemented mini-SOC system works as intended, several controlled simulations were performed within the home network. The purpose of these tests was to observe how the detection modules respond to suspicious behavior and whether the correlation mechanism increases alert reliability.

In the first scenario, a port scan was simulated using the nmap tool from one client device in the same network. The scan targeted multiple ports on another device connected to the managed switch. As expected, the port scan detection module identified a sequence of rapid connection attempts coming from the same source IP address. The system classified this behavior according to the predefined severity model and generated a structured alert entry.

```
[[redacted]@Jamess-Air ~ % sudo nmap -p 22,80 192.168.0.41
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-25 21:54 +0100
Nmap scan report for soc (192.168.0.41)
Host is up (0.0091s latency).

PORT   STATE  SERVICE
22/tcp open   ssh
80/tcp closed http
MAC Address: 88:A2:9E:84:BD:25 (Raspberry Pi (Trading))

Nmap done: 1 IP address (1 host up) scanned in 1.07 seconds
```

Figur 3: Simulated port scan using nmap

```
2026-02-25 21:53:47 [HIGH] port_probe src=192.168.0.40 dst=192.168.0.41 dpt=3389
2026-02-25 21:53:49 [HIGH] port_probe src=192.168.0.40 dst=192.168.0.41 dpt=445
2026-02-25 21:53:50 [INFO] dns_live rate_per_min=54 threshold=250 iface=eth0
2026-02-25 21:54:00 [INFO] dns_live rate_per_min=6 threshold=250 iface=eth0
2026-02-25 21:54:08 [INFO] port_probe src=192.168.0.40 dst=192.168.0.41 dpt=22
2026-02-25 21:54:08 [INFO] port_probe src=192.168.0.40 dst=192.168.0.41 dpt=80
2026-02-25 21:54:10 [INFO] dns_live rate_per_min=54 threshold=250 iface=eth0
2026-02-25 21:54:20 [INFO] dns_live rate_per_min=30 threshold=250 iface=eth0
2026-02-25 21:54:30 [INFO] dns_live rate_per_min=30 threshold=250 iface=eth0
```

Figur 4: Port scan detection event (Raspberry Pi)

In the second scenario, abnormal DNS activity was simulated by generating a large number of DNS queries within a short time period. This behavior exceeded the configured threshold values and triggered the DNS anomaly detection module. The event was recorded in JSON format and assigned a severity level based on the intensity of the activity. This test confirmed that the system can distinguish between normal browsing behavior and unusually high DNS activity.

```
2026-02-25 21:59:07 [INFO] dns_live rate_per_min=6 threshold=250 iface=eth0
2026-02-25 21:59:17 [INFO] dns_live rate_per_min=30 threshold=250 iface=eth0
2026-02-25 21:59:27 [INFO] dns_live rate_per_min=6 threshold=250 iface=eth0
2026-02-25 21:59:37 [INFO] dns_live rate_per_min=6 threshold=250 iface=eth0
2026-02-25 21:59:47 [INFO] dns_live rate_per_min=66 threshold=250 iface=eth0
2026-02-25 21:59:57 [INFO] dns_live rate_per_min=72 threshold=250 iface=eth0
2026-02-25 22:00:07 [MEDIUM] dns_live rate_per_min=132 threshold=250 iface=eth0
2026-02-25 22:00:17 [INFO] dns_live rate_per_min=48 threshold=250 iface=eth0
2026-02-25 22:00:27 [INFO] dns_live rate_per_min=78 threshold=250 iface=eth0
```

Figur 5: DNS anomaly detection triggered by high query rate (Raspberry Pi)

To evaluate the correlation logic, both port scanning and abnormal DNS activity were triggered from the same device within the defined time window. In this case, the system elevated the severity level to CRITICAL.

```
2026-02-25 22:03:18 [INFO] dns_live rate_per_min=30 threshold=250 iface=eth0
2026-02-25 22:03:28 [INFO] dns_live rate_per_min=24 threshold=250 iface=eth0
2026-02-25 22:03:38 [MEDIUM] dns_live rate_per_min=162 threshold=250 iface=eth0
2026-02-25 22:03:48 [INFO] dns_live rate_per_min=60 threshold=250 iface=eth0
2026-02-25 22:03:58 [INFO] dns_live rate_per_min=30 threshold=250 iface=eth0
2026-02-25 22:04:08 [CRITICAL] dns_live rate_per_min=9318 threshold=250 iface=eth0
2026-02-25 22:04:20 [MEDIUM] dns_live rate_per_min=228 threshold=250 iface=eth0
2026-02-25 22:04:22 [CRITICAL] port_probe src=192.168.0.40 dst=192.168.0.41 dpt=23
2026-02-25 22:04:22 [CRITICAL] correlation src=192.168.0.40 dst=192.168.0.41 dpt=23 dns_rate=9318 reason=dns_plus_port_window=60s
```

Figur 6: Correlated DNS and port scan activity resulting in CRITICAL alert (Raspberry Pi)

This demonstrated that the correlation engine successfully combines multiple suspicious indicators and increases confidence in the alert. Instead of treating events separately, the system evaluates them together to produce a more meaningful result.

All detected events were logged in structured files and displayed on the terminal-based dashboard in real time. High-severity events, especially those marked as **CRITICAL** by the correlation engine, automatically triggered email notifications through the configured SMTP service. This allowed important security incidents to be reported even when the dashboard was not being watched. The successful delivery of these emails showed that the notification system worked reliably during the simulated attack scenarios.

Figur 7: Email notification generated for a CRITICAL security event (Gmail)

Overall, the results show that a low-cost hardware platform can effectively detect anomalous behavior and apply event correlation within a home network environment.

Discussion

The aim of the project was to investigate whether it is possible to build an economically feasible and functional network security system for home networks without the hardware and service resources of corporate companies. As seen at the end of the project, it is possible to establish a cost-effective and fully flexible network monitoring and alert system. The Raspberry Pi used in the project successfully detected port scanning and DNS anomalies, which are among the most commonly used techniques by attackers, and generated alerts for the user. In addition, these alerts were not evaluated only as isolated activities, but were also correlated, and when a pattern was detected, the system generated a higher-severity alert.

This network monitoring system is simple and transparent. The logic for detecting port scans and DNS anomalies is completely based on predefined rules. For individuals connected to home networks who do not have expertise in information systems, this provides added value. Compared to sophisticated systems, it is more understandable and can be modified according to specific needs. In addition, it is highly efficient in terms of cost. Especially the latest model, Raspberry Pi 5, is relatively low cost but provides performance close to a standard computer. This makes it sufficient for running a network monitoring system, storing logs, and sending alerts. This finding

directly answers one of the main research questions of the project: whether it is possible to protect a home network without professional and expensive systems. Furthermore, it was observed that performing correlation, in addition to anomaly detection and alert generation, increases alert quality and reduces unnecessary noise.

On the other hand, although the simplicity of the system is an advantage, it is based on predefined threshold values, and therefore it cannot be fully trusted to detect advanced attack techniques at all times. The system does not include sophisticated features such as signature-based analysis that are used in professional IDS solutions.

As a result, this system cannot be compared to the network monitoring systems or SOC units of corporate organizations. However, through network monitoring, anomaly detection, event correlation, and alert generation, it has successfully increased security visibility in small home network environments.

Conclusion

In this project, a low-cost mini-SOC system was designed, implemented, and tested to provide a certain level of security for small home networks. Although this system does not evaluate multiple anomalies simultaneously or perform deep correlation like advanced enterprise monitoring systems, it can detect port scans and DNS anomalies and, when these are detected together, elevate the alert severity from “HIGH” to “CRITICAL”.

Based on the performance observed during testing, the project demonstrates that security solutions can be implemented even on low-performance devices compared to enterprise systems.

In conclusion, this project shows that a simple and small-scale security architecture can significantly increase security visibility in home network environments.

Ethical Analysis

Although this project aims to provide network security by analyzing network traffic, these activities also have an ethical dimension. The traffic analysis performed in this project does not inspect the content of network communications. However, since DNS queries can be monitored, it is possible to see which websites users attempt to access.

As mentioned at the beginning of the report, network traffic is monitored in corporate environments under specific contracts and organizational policies, and employees are informed about such monitoring activities. In a home environment, however, once professional life ends and private life begins, an ethical dimension emerges.

Within the scope of this project, individuals connected to the home network were informed about the project, including what would be monitored, what information could be visible, and what would remain private. Their consent was obtained before the monitoring system was activated. Furthermore, the collected monitoring data was not shared with any third party.